Assignment 4

1. **[4+3+3 marks]** The Dorben Electronics Co., manufacturer of resistors, screens potential quality control inspectors before they are hired. Dorben has developed the following preemployment test. Each potential employee is presented with the same set of 1,000 resistors of which 500 are defective. The results for two applicants are as follows:
(1) of the 500 good resistors, applicant 1 labeled 100 as defective, and of the 500 bad resistors, applicant 1 labeled 200 as defective; (2) of the 500 good resistors, applicant 2 labeled 50 as defective, and of the 500 bad resistors, applicant 2 labeled 300 as defective.

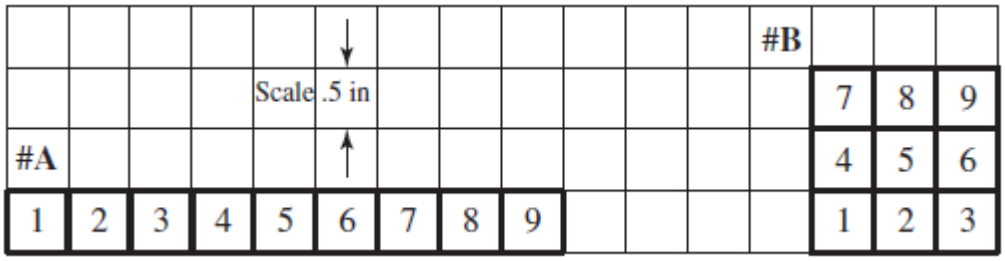
(a) Treat picking a defective resistor as defective as a hit. Fill in the following table.

	Applicant 1	Applicant 2
Hit rate		
False-alarm rate		
Miss rate		
Corrective rejection rate		
d'		

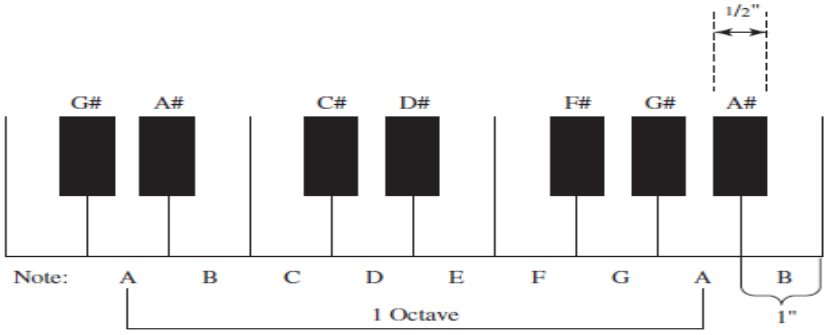
(b) Given that Dorben places a very large emphasis on quality control (i.e., they don't wish to sell a defective product at any cost), which applicant would better fulfill its goal? Why?

(c) Given that Dorben wishes to hire the most efficient inspector (i.e., most correct), which applicant would the company hire?

2. [**5 marks**] Disregarding the digit 0, which keyboard is fastest for entering digits using one finger? Assume that the home position is digit 5. (Hint: Calculate the index of difficulty.)



3. [**3+2 marks**] Pianists must often hit keys in very quick succession. The figure below displays a typical piano keyboard.

(a) Compare the index of difficulty for striking C, C#, F, and F#. Assume that one is starting from the A key and distances are measured center to center.

(b) If starting from the A key, it takes 200 ms to strike the C key and 500 ms to strike the F key, what is the bandwidth of a typical pianist?